

YASIR KHAN

Senior Mobile Application Developer | iOS Native (Swift) | Android (Kotlin) | Flutter (Dart)

Mardan, Khyber Pakhtunkhwa, Pakistan | ykhkpk@gmail.com | +92 315 0012413 | linkedin.com/in/yasir-khan-mobile | Portfolio: ykhkpk.github.io

PROFESSIONAL SUMMARY

Senior Mobile Application Developer with **5+ years of experience** architecting, developing, and deploying high-performance **iOS Native, Android Native, and Flutter** applications from conception to App Store and Google Play production. Specialist in **Modern Clean Architecture, strict SOLID principles**, protocol-driven decoupled architectures, and reactive concurrency. Deep expertise in **Swift 6, SwiftUI, SwiftData, GRDB (SQLite), CoreText, PDFKit, CryptoKit**, Kotlin, Jetpack Compose, Hilt, Room, and Flutter/Riverpod. Proven track record delivering **14+ commercial applications** spanning vector PDF document editors, offline-first peer-to-peer encrypted mesh networks, generative AI/LLM assistants, multimedia download engines, and Islamic digital tools. Skilled engineering leader with proven experience driving technical design decisions, client communications, and mentoring junior engineers.

CORE HIGHLIGHTS

- 5+ Years Experience | 14+ Apps Delivered** to Production
- iOS Native Expert:** Swift 6, SwiftUI, SwiftData, GRDB, PDFKit
- Offline-First & Reactive:** GRDB DatabasePool, Room, SwiftData Actors
- AI & Vision Integration:** Apple Vision OCR, Streaming LLM Chat, STT
- Modern Clean Architecture & SOLID Principles** (Domain/Data/Presentation)
- Android Native Expert:** Kotlin, Jetpack Compose, Coroutines, Flow
- P2P Mesh & Cryptography:** MultipeerConnectivity, CryptoKit, WebRTC
- Engineering Leadership:** Architecture Decisions, Code Reviews, Agile

PROFESSIONAL EXPERIENCE

Mobile Application Developer (Lead Mobile Architect) May 2022 – Present
Ainsoftech — Lahore, Pakistan

PDFEditor (Native iOS 18+ — Swift 6, SwiftUI, CoreText, PDFKit, SwiftData)

- Architected an enterprise vector PDF editor using **Modern Clean Architecture** (Domain, Data, Presentation, Core) and strict **SOLID principles** with protocol-based dependency inversion, eliminating layer coupling.
- Pioneered true in-place PDF text and graphics editing via low-level **PDFKit, CoreText**, and **CoreGraphics** page rendering, including surgical background raster erasure to prevent ghosting during text mutation.
- Integrated **Apple Vision OCR** for text recognition on scanned documents, lossless multi-tier PDF compression, and Command-Pattern Undo/Redo historical state management with SwiftData transaction persistence.

AetherChat (Native iOS — Swift 6, SwiftUI, SwiftData, MultipeerConnectivity, CryptoKit)

- Engineered an offline-first, decentralized peer-to-peer mesh messaging system operating over **MultipeerConnectivity** (Wi-Fi & Bluetooth) without internet infrastructure, adhering to SOLID single-responsibility protocols.
- Implemented end-to-end cryptographic identity using hardware-backed **Ed25519** digital signatures, **Curve25519 (X25519)** key exchange stored in Apple Keychain, and **ChaCha20-Poly1305** authenticated payload encryption.
- Built real-time peer-to-peer **WebRTC calling**, voice messaging with live **AVAudioSession** waveform metering, multi-megabit Wi-Fi file streaming, and transactional data synchronization via SwiftData `@ModelActor` with zero data races.

QuickSnap Video (Native iOS — SwiftUI, App Intents, Background URLSession, PhotoKit)

- Developed a high-speed video ingestion utility integrated with iOS system ergonomics via **App Intents, Action Button**, and iPhone **Back Tap** accessibility triggers with automatic clipboard URL detection.
- Engineered out-of-process background download pipelines using **URLSession background configuration (nsurlsessionond)** and direct **PhotoKit (PHPhotoLibrary)** Camera Roll export, ensuring uninterrupted downloads during app suspension.

Quran Kareem (Native iOS & Android Native — Swift 6, SwiftUI, GRDB, Kotlin, Jetpack Compose, Clean Architecture)

- Architected full native **iOS** and **Android** apps employing **Modern Clean Architecture** (Domain, Data, Presentation) and **SOLID principles** (Interface Segregation with segregated repository protocols, Dependency Inversion with decoupled Use Cases).
- Engineered high-performance offline-first local storage on iOS using **GRDB (SQLite)** with DatabasePool multi-reader concurrency, type-safe **PersistableRecord** models, and reactive **AsyncStream** database observation for real-time bookmark and reading state synchronization; Room database on Android.
- Implemented a sensor-fused real-time Qibla compass (CoreLocation & Rotation Vector with GeomagneticField correction), worldwide prayer calculation engine, and a 14-line Mushaf reader with background audio streaming and active Ayah synchronization.

Stylo AI & AI Prompt Suite (Flutter — Riverpod, Large Language Models, Speech-to-Text)

- Engineered AI outfit recommendation app parsing unstructured LLM outputs into typed Dart models with custom JSON boundary extraction, factoring in weather telemetry and occasion detection.
- Developed an AI prompt toolkit with real-time token-streaming LLM chat and low-latency microphone audio streaming for live speech-to-text transcription.

Scoop Messenger & Engineering Leadership (Android Native, WebSockets & Technical Mentorship)

- Built real-time messaging, audio/video calling, and media sharing backed by persistent foreground services with exponential-backoff auto-reconnection achieving 99.5% connection uptime.
- Mentored junior iOS, Android, and Flutter developers; established code review standards, SOLID design principles, sprint roadmaps, and client communication.

PROFESSIONAL EXPERIENCE (CONTINUED)

Mobile Developer

CodingStorms Technology — Islamabad, Pakistan

January 2022 – May 2022

- Developed native Android and cross-platform Flutter applications for enterprise and startup clients.
- Integrated Firebase Authentication, Cloud Firestore, Cloud Storage, RESTful APIs, and mobile payment gateways.
- Translated intricate Figma design systems into pixel-perfect, responsive mobile layouts adhering to strict UI/UX guidelines.
- Collaborated closely within Agile Scrum sprints delivering production-ready releases ahead of schedule.

Junior Mobile Developer

Solution of Technologies — Islamabad, Pakistan

2020 – 2021

- Developed core Android applications using Java, Kotlin, XML layouts, and ViewBinding.
- Implemented MVVM architecture patterns, Retrofit networking modules, Room database persistence, and Firebase integration.
- Participated in daily standups, code reviews, quality assurance bug triaging, and deployment preparation.

KEY PROJECTS PORTFOLIO

PDFEditor (iOS Native — Swift 6, SwiftUI, PDFKit, CoreText, Vision OCR, SwiftData):

Full-featured vector PDF document editor featuring true in-place content editing of existing text, fonts, colors, and graphics with CoreText reconstruction, background erasure, Vision OCR, and multi-version snapshots.

QuickSnap Video (iOS Native — SwiftUI, App Intents, Background URLSession, PhotoKit):

Automated video download engine integrated with Apple Action Button and Back Tap accessibility triggers via App Intents, utilizing background nsurlsessiond for out-of-process downloads directly to Camera Roll.

Scoop Messenger (Android Native — WebSockets, WebRTC, Foreground Services):

Real-time communication suite featuring audio/video calling, group messaging, and persistent background connection management with 99.5% uptime.

AI Prompt App (Flutter — Riverpod, LLM Streaming, Speech-to-Text):

AI-powered prompt generator featuring real-time streaming LLM chat and low-latency microphone audio streaming feeding live speech-to-text transcription.

AetherChat (iOS Native — Swift 6, SwiftUI, SwiftData, MultipeerConnectivity, CryptoKit):

Decentralized, offline-first peer-to-peer messenger operating over local mesh networks without internet. Features Ed25519 & Curve25519 cryptographic security, WebRTC calling, voice notes with live waveform metering, and glassmorphic UI.

Quran Kareem (Native iOS & Android — Swift 6, SwiftUI, GRDB, Kotlin, Jetpack Compose, Clean Arch):

Offline-first Islamic companion for iOS and Android built with Clean Architecture and SOLID design. Features high-performance GRDB (SQLite) on iOS with reactive AsyncStream observation, sensor-fused Qibla Compass, worldwide prayer calculation, and 14-line Mushaf reader with audio sync.

Stylo AI (Flutter — Riverpod, Generative AI, Multi-Modal Image Analysis):

AI fashion recommendation platform using structured LLM response extraction, weather data telemetry, and occasion detection for personalized style suggestions.

EasySkills LMS (Flutter — Riverpod, Clean Architecture, LMS / EdTech):

Learning Management System featuring passwordless 6-digit OTP authentication, Bunny.net CDN video streaming, creator course tools, and offline lesson caching.

TECHNICAL SKILLS

Languages: Swift 6, Kotlin, Dart, Java, C/C++ (Basic)

iOS Native: SwiftUI, UIKit, iOS SDK (18+), SwiftData, GRDB (SQLite), PDFKit, CoreText, CoreGraphics, Apple Vision Framework, MultipeerConnectivity, CryptoKit, App Intents, AVAudioSession, PhotoKit, Background URLSession, Keychain Services, XCTest

Android Native: Kotlin, Jetpack Compose, Compose Multiplatform, Hilt, Coroutines, Flow, StateFlow, Navigation Component, Room Database, WorkManager, DataStore, CameraX, Sensors API, Canvas 2D, Google Maps SDK

Flutter & Cross-Platform: Flutter SDK, Riverpod (2.6+), Provider, Platform Channels, Custom Painters, Dart Concurrency

Architecture & Design: Modern Clean Architecture (Domain/Data/Presentation/Core), Strict SOLID Principles (Interface Segregation, Dependency Inversion), MVVM, MVI, Repository Pattern, Command Pattern (Undo/Redo), Swift 6 Strict Concurrency (Actors, Sendable, ModelActor)

AI, ML & Real-Time: Apple Vision OCR, LLM Integration (OpenAI, Gemini), Streaming Token APIs, Real-Time Speech-to-Text, Audio Streaming Pipeline, Prompt Engineering, Google ML Kit, WebRTC, WebSockets

Backend, Storage & Cloud: GRDB (SQLite), SwiftData, Room Database, SQLite, REST APIs, Retrofit, Ktor, URLSession, Firebase (Auth, Firestore, Cloud Storage, FCM, Crashlytics)

DevOps & Tooling: Xcode, Android Studio, VS Code, Git, GitHub, TestFlight, App Store Connect, Google Play Console, CI/CD Pipelines, Agile/Scrum

EDUCATION

Bachelor of Science in Computer Science (BSCS)

Sarhad University of Science and Information Technology — Peshawar, Pakistan

Graduated